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Vox2Face: Speech-Driven Face Generation via Identity-Space Alignment and Diffusion Self-Consistency

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Abstract

Speech-driven face generation aims to synthesize a face image that matches a speaker's identity from speech alone. However, existing methods typically trade identity fidelity for visual quality and rely on large end-to-end generators that are difficult to train and tune. We propose Vox2Face, a speech-driven face generation framework centered on an explicit identity space rather than direct speech-to-image mapping. A pretrained speaker encoder first extracts speech embeddings, which are distilled and metric-aligned to the ArcFace hyperspherical identity space, transforming cross-modal regression into a geometrically interpretable speech-to-identity alignment problem. On this unified identity representation, we reused an identity-conditioned diffusion model as the generative backbone and synthesized diverse, high-resolution faces in the Stable Diffusion latent space. To better exploit this prior, we introduce a discriminator-free diffusion self-consistency loss that treats denoising residuals as an implicit critique of speech-predicted identity embeddings and updates only the speech-to-identity mapping and lightweight LoRA adapters, encouraging speech-derived identities to lie on the high-probability identity manifold of the diffusion model. Experiments on the HQ-VoxCeleb dataset show that Vox2Face improves the ArcFace cosine similarity from 0.295 to 0.322, boosts R@10 retrieval accuracy from 29.8% to 32.1%, and raises the VGGFace Score from 18.82 to 23.21 over a strong diffusion baseline. These results indicate that aligning speech to a unified identity space and reusing a strong identity-conditioned diffusion prior is an effective method to jointly improve identity fidelity and visual quality.

Keywords: speech-driven face generation; cross-modal generation; face prior; diffusion self-consistency

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1. Introduction

Humans are to some extent able to "hear a face and see a voice." Studies in psychology and cognitive science have shown that people can match unfamiliar voices to faces above chance in cross-modal identity tasks. This implies that the voice and face rely on a shared latent identity space [1]. Behavioral and neuroimaging studies further indicate that faces and voices are processed along parallel yet interacting pathways in perception and the brain, providing a biological and cognitive basis for predicting facial attributes—and even

identity consistency—from speech alone. In computer vision and multimodal learning, cross-modal biometric matching and retrieval have become active research areas, including tasks such as selecting the correct face for a given voice from a gallery of candidates [2].

Therefore generating or retrieving an identity-consistent face from speech alone is a natural next step in studying voice–face correspondence. It offers a concrete testbed for probing how shared identity information is distributed across modalities, while also supporting applications such as personalized avatar creation, identity-aware multimodal retrieval, and forensic assistance under limited visual evidence [2–4]. At the same time, reconstructing faces from speech inevitably raises concerns about privacy, bias amplification, and potential misuse, which calls for techniques that are not only accurate but also controllable and compliant by design.

In recent years, the rapid progress of diffusion-based image generators has made such goals more realistic. Denoising Diffusion Probabilistic Models (DDPMs) and Latent Diffusion Models (LDMs) now provide high-fidelity, controllable synthesis at scale, and Stable Diffusion–style architectures have become standard backbones for conditional image generation [5–7]. In the face domain, identity-conditioned variants such as Arc2Face demonstrate that a pretrained diffusion model can be steered by compact identity embeddings (e.g., ArcFace) to produce diverse yet ID-consistent faces, effectively serving as a powerful prior for identity-driven generation [8]. This suggests that, instead of training a large speech-to-image generator end-to-end, one can factor the problem into (i) mapping speech into a well-structured identity space and (ii) leveraging a strong, frozen identity-conditioned diffusion prior on top of that space—which is precisely the perspective adopted in this work.

However, directly training an end-to-end speech-to-face generator remains challenging. First, speech embeddings encode not only speaker identity but also linguistic content, emotional state, prosody, and recording conditions, which introduces substantial cross-modal mismatch between speech and facial identity features and makes end-to-end training unstable and sensitive to distribution shifts. Second, high-quality paired audio–face datasets with broad coverage are not widely available at scale, which complicates training large generators from scratch to jointly learn robust cross-modal alignment and high-fidelity rendering. While the importance of identity-related feature alignment has been discussed in prior work (e.g., [9]), our approach operationalizes this direction through ArcFace-supervised identity distillation and a frozen identity-conditioned diffusion prior. Third, existing methods typically face a pronounced trade-off between identity preservation and visual realism: improving identity discriminability tends to reduce sample diversity or perceptual quality, whereas pushing for more natural, photorealistic renderings can weaken identity constraints and harm cross-domain generalization and practical usability.

To address these issues, we propose Vox2Face, a speech-driven face generation framework explicitly centered on identity metrics rather than direct speech-to-image regression. First, we mitigate the modality gap and the noise sensitivity of speech embeddings by aligning the output of a pretrained speaker encoder to the hyperspherical identity space of a pretrained face recognizer (ArcFace) via knowledge distillation and metric learning. This converts the original high-dimensional, content-entangled mapping into a geometrically well-posed speech-to-identity alignment problem in a unified space. Second, to overcome the lack of large paired speech–face datasets and the instability of training huge generators end-to-end, we reuse a pretrained identity-conditioned diffusion model (Arc2Face) as a frozen generative prior: faces are synthesized directly from the aligned identity embeddings, so the model benefits from a strong face prior without retraining the diffusion backbone. Third, to reduce the residual trade-off between identity fidelity and visual quality, we exploit the diffusion model's own denoising process to construct a diffusion

self-consistency training mechanism: instead of adding an extra discriminator, we treat denoising residuals as an implicit supervision signal for the speech-to-identity mapping and update only the speech-to-identity network and the LoRA adapters injected into the generator. This encourages speech-derived identity embeddings to stay close both to the ground-truth ArcFace codes and to the high-probability identity manifold of the diffusion model, thereby improving identity preservation and perceptual quality in a unified way.

We design Vox2Face around identity-centric speech-driven generation, with contributions at the levels of representation alignment, generative priors, and training mechanisms:

(1) a unified identity-space cross-modal alignment framework on the ArcFace unit hypersphere;
(2) a speech-driven face generation scheme built on a frozen identity-conditioned diffusion prior with lightweight LoRA adaptation; and
(3) a unified objective that combines ArcFace-supervised cross-modal identity alignment with an SDS-style diffusion self-consistency loss in a two-stage training regime.

Experiments on HQ-VoxCeleb demonstrate that this design significantly outperforms representative baselines in identity similarity, speech-to-face retrieval, and perceptual quality metrics, validating the effectiveness of "align identity space first, then reuse a strong identity-conditioned diffusion prior" for speech-driven face generation.

2. Related Work

Speech-driven face generation. Speech-driven face generation aims to synthesize a face image that is consistent with a speaker's identity from speech alone, making it a highly challenging cross-modal generation problem. Early work typically adopts convolutional or autoencoder-style architectures that extract speech features, regress them to a face representation, and then decode an image. Speech2Face learns a mapping from speech to the intermediate feature space of a pretrained face recognition network using millions of talking-head videos as weak supervision, then reconstructs static face images from these features, demonstrating that attributes such as gender, age, and ethnicity are statistically correlated across voice and face [9]. Wav2Pix instead emphasizes fully end-to-end modeling: raw waveforms are fed into a convolutional network and a conditional GAN to directly generate face images from audio–visual co-occurrence, without explicit attribute labels or additional identity priors [10]. Subsequent GAN- and encoder–decoder-based methods further incorporate facial priors, attention mechanisms, and L1/L2/cosine losses, typically first mapping speech into high-level latent representations and then decoding them into images, in an effort to preserve as much identity- and attribute-related information as possible [11–14].

With the rise of diffusion models in image synthesis, recent approaches have begun to replace GANs with diffusion-based generators for speech-driven face generation. A representative example is the Speech-Conditioned Latent Diffusion Model (SCLDM), which injects dual speech and face encoders into an LDM, uses contrastive pre-training to align speaker attributes, and introduces a statistical face prior into the diffusion process to obtain more realistic and identity-preserving faces [15]. Empirical results across these diffusion-based approaches indicate that, given a powerful generative backbone and a reasonable voice–face alignment strategy, speech-to-face models can substantially outperform traditional GAN-based methods in visual fidelity and training stability [16,17]. Sound2Vision [18] also demonstrates diffusion-based audio-conditioned generation on high-quality face video data (CelebV-HQ), but it primarily targets open-domain audio–visual latent alignment rather than identity-preserving speech-to-face generation. At the same time, within identity-preserving speech-driven face generation, existing diffusion-based methods still often rely on end-to-end training of large generators and directly

condition the generator on speech features, which leaves modality mismatch and the limited availability of curated, identity-clean paired speech–face supervision as open issues.

Cross-modal representation learning. In audio–visual self-supervised learning, L3-Net (Look, Listen and Learn) learns a joint audio–visual embedding space from unlabeled videos under the weak assumption that the sound and the visual frame from the same clip should correspond, showing that transferable multimodal representations can be obtained without explicit annotations [19]. SoundNet adopts a student–teacher framework on roughly two million unlabeled videos, transferring discriminative power from pretrained visual recognition networks to the acoustic modality and achieving strong performance on multiple audio benchmarks [20]. At a finer-grained "voice–face" identity level, Nagrani et al. learn a cross-modal metric space directly on paired face–voice data, enabling tasks such as "given a voice, select the corresponding face from a gallery," and report performance substantially above chance—and in some settings approaching or even exceeding human accuracy—on in-the-wild datasets such as VoxCeleb and VGGFace [2]. These works provide mature baselines for measuring and evaluating cross-modal identity consistency.

Knowledge distillation is a standard mechanism for cross-modal alignment: a strong "teacher" encoder (e.g., an image or face encoder) is frozen, and its outputs are used as soft targets to constrain the "student" representations of another modality (e.g., speech), encouraging the student to converge to the teacher's semantic geometry [20]. In practice, this can be implemented via MSE or cosine losses for direct feature matching, and can be further strengthened with temperature-scaled soft targets and contrastive objectives. Crucially, the choice of teacher space determines the nature of the alignment target. General-purpose semantic cross-modal spaces such as CLIP, trained by image–text contrastive learning, provide broad semantic transfer and are well suited for modeling text–image semantic consistency [21]. In contrast, face recognition spaces such as ArcFace, trained with additive angular margin losses on the unit hypersphere, explicitly enlarge inter-class angular margins and yield a geometrically well-structured metric space centered on identity discrimination [22]. Such identity-focused spaces have been shown, for example by Arc2Face, to support high-quality, identity-consistent face generation when used as the sole conditioning signal for a diffusion model [8]. Our work follows this line by choosing an identity-focused teacher (ArcFace) and distilling speech embeddings into its hyperspherical space before generation.

Image generation and conditional diffusion models. Diffusion models have become the dominant paradigm for high-quality image synthesis. DDPMs formulate generation as a Markov chain of gradual noising and denoising, sampling images from Gaussian noise and achieving performance comparable to or better than GANs on many benchmarks [23]. LDMs move this process from pixel space to the latent space of a pretrained autoencoder, reducing computational cost while preserving visual fidelity; via cross-attention, they naturally incorporate text and other conditioning signals, and have become a standard backbone for text-to-image and multimodal generation [5]. Building on this line, Imagen demonstrates that combining large language models with diffusion models enables photorealistic text-to-image generation with strong language understanding [24].

For conditional control and personalization, recent work typically freezes the large pretrained backbone and modifies only the conditioning pathway or a small set of parameters. ControlNet keeps the base Stable Diffusion weights fixed and adds an auxiliary network to inject structural conditions such as edges, depth, and pose, enabling spatially controllable generation [6]. IP-Adapter introduces a lightweight adapter that feeds image prompts into pretrained text-to-image diffusion models, supporting joint image–text conditioning without retraining the backbone [7]. For identity conditioning, methods such as InstantID, ID-Booth, and PhotoMaker augment the model with dedicated identity branches or

stacked identity embeddings to achieve personalized subject synthesis, maintaining identity consistency while offering flexible control over style and pose under a "freeze Stable Diffusion/SDXL, adapt a few modules" paradigm [25–27]. Arc2Face further pushes this direction by using ArcFace identity vectors as the sole conditioning signal and fine-tuning Stable Diffusion on large-scale face recognition datasets such as WebFace42M, yielding an identity-conditioned foundation model that can generate diverse, highly identity-consistent faces given only an ArcFace embedding [8].

Score Distillation Sampling (SDS) complements these design patterns by treating frozen diffusion models as priors and critics. DreamFusion optimizes 3D scene parameters (e.g., NeRF) by matching the score of a text-conditioned diffusion model in parameter space, enabling text-to-3D generation without any 3D supervision [28]. Follow-up work such as debiased SDS (D-SDS) improves SDS in terms of bias and stability, showing that even without fine-tuning the diffusion backbone, its denoising score alone can effectively guide optimization in external parameter spaces [29]. Overall, this line of research outlines a trajectory that couples (i) strong pretrained diffusion backbones, (ii) lightweight conditioning and identity-personalization modules, and (iii) SDS-style distillation mechanisms. However, in speech-driven face generation, existing methods still tend to either train speech-conditioned generators from scratch or use diffusion models only as generic differentiable decoders. They rarely combine a pretrained identity-conditioned diffusion model with an SDS-style critic to explicitly constrain speech-derived representations to the intersection of a pretrained identity manifold and the generator's data manifold—a gap that our identity-aligned, diffusion self-consistent Vox2Face framework is designed to address.

3. Preliminaries

In this section, we briefly introduce the identity representation space, the pretrained identity-conditioned generator, and the speech-to-identity mapping used in our framework, which together prepare for the unified optimization objective and training scheme described in Section 4.

3.1. ArcFace Identity Space

Let X denote the face image domain, and let the identity embedding space be the unit hypersphere

$$S^{d-1} = \left\{ z \in R^d \mid \|z\|_2 = 1 \right\} \tag{1}$$

ArcFace learns discriminative face representations on S^{d-1} using an additive angular margin loss. After ℓ_2-normalizing both feature vectors and classifier weights, the logit for the target class replaces $\cos\theta_y$ with $\cos(\theta_y + m)$, where θ_y is the angle between the feature and the target-class weight, and m is a fixed angular margin. This explicitly enlarges inter-class margins in terms of geodesic distance on the sphere and provides clearer geometric interpretation and stronger identity discriminability than standard softmax or other margin-based variants [22]. We denote ArcFace identity embeddings by

$$z_{arc} \in R^{512}, \ \|z_{arc}\|_2 = 1 \tag{2}$$

and use cosine similarity (i.e., inner product on the sphere) as our primary measure of identity consistency. In the following, this hyperspherical ArcFace space will serve as a shared identity representation space for both speech and face modalities in Vox2Face.

3.2. Arc2Face Identity-Conditioned Face Generator

Arc2Face is an identity-conditioned face generation foundation model built on Stable Diffusion. Given an ArcFace embedding z_{arc} for any individual, the model can synthesize diverse, highly identity-consistent face images within a few seconds. Its backbone is Stable Diffusion v1.5. First, the ArcFace vector $z_{arc} \in R^{512}$ is mapped by a projection network

$$P(\cdot) : R^{512} \rightarrow R^{d_{cond}} \tag{3}$$

to obtain an identity condition vector $e_{ID} = P(z_{arc})$. This vector is then injected into the cross-attention layers of the UNet in the latent diffusion model, replacing or complementing the original text condition so that the denoising network is guided by the identity vector during sampling. As a result, the model can generate varied faces that share the same identity, even without any textual prompt.

During training, Arc2Face fine-tunes Stable Diffusion on high-quality, upsampled WebFace42M images under ID conditioning, significantly improving identity coverage and similarity. In our work, we keep all backbone weights of Arc2Face frozen and treat it as a pretrained identity-conditioned generative prior and differentiable critic. Task-specific adaptation is performed only via a small number of LoRA modules injected into the text encoder and the conditional branches of the UNet [8].

3.3. Mapping Speech to Identity Embeddings

Let $v_{spk} \in R^{d_{spk}}$ denote the speaker embedding extracted from a speech segment by a pretrained speech encoder. One of our core goals is to learn a mapping

$$g_\phi : R^{d_{spk}} \rightarrow S^{\mathrm{d}-1},\ \hat{z}_{arc} = g_\phi\left(v_{spk}\right) \tag{4}$$

such that the predicted identity vector $\hat{z}_{arc}$ is (i) geometrically aligned with the ground-truth ArcFace embedding of the corresponding face on the hypersphere, and (ii) lies in a region where Arc2Face can reliably generate high-quality, natural faces.

In later sections, we jointly optimize g_ϕ and the LoRA adaptation parameters using ArcFace-based identity supervision and a diffusion self-consistency constraint, thereby achieving end-to-end modeling from speech to identity-conditioned face generation without modifying the large diffusion backbone.

4. Method

4.1. Overall Framework

Our method adopts an end-to-end design in which a single identity vector is propagated through the entire pipeline (Figure 1). Starting from a speech input, the model never regresses directly from speech to images or generic latent codes; instead, it operates explicitly on this identity vector. Given a speech segment a, the pipeline computes

$$\mathrm{v_{spk}} = \mathrm{f_{spk}(a)},\ \hat{\mathrm{z}}_{\mathrm{arc}} = \mathrm{g}_\phi\left(\mathrm{v_{spk}}\right),\ \hat{\mathrm{x}} = \mathrm{G}_\theta^{\mathrm{Arc2Face}}(\hat{\mathrm{z}}_{\mathrm{arc}}, \xi) \tag{5}$$

where f_{spk} is a pretrained speaker encoder, g_ϕ is the speech-to-ArcFace identity mapping network, $G_\theta^{Arc2Face}$ is the frozen Arc2Face identity-conditioned diffusion generator, and ξ denotes sampling noise.

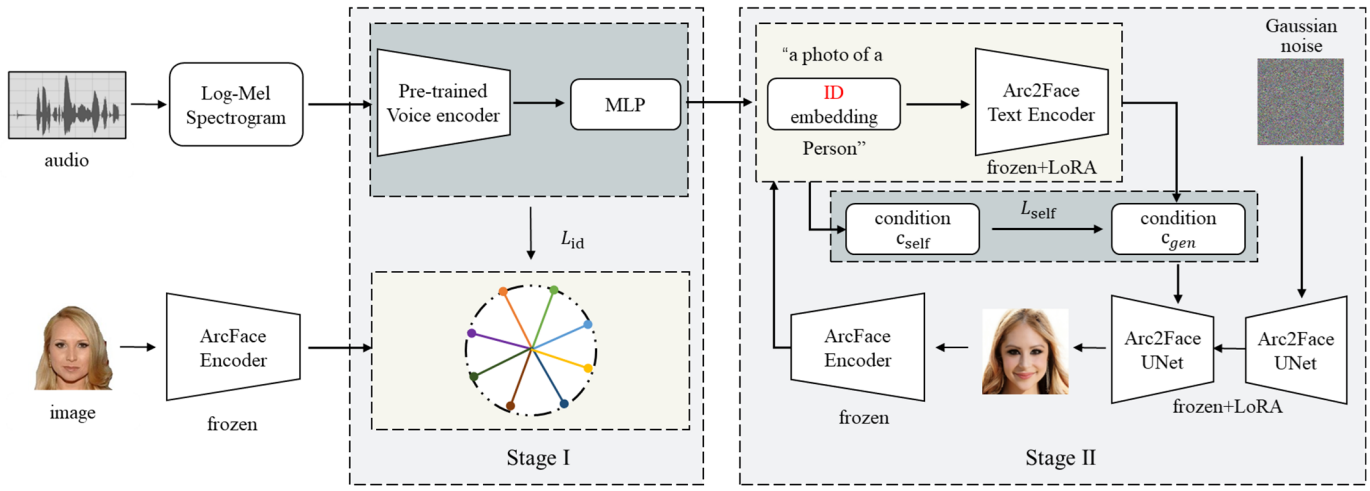

Figure 1. Overview of the proposed Vox2Face framework. Stage I aligns speech embeddings to the ArcFace identity hypersphere via a pretrained speaker encoder and an MLP mapping network. Stage II reuses the frozen Arc2Face identity-conditioned diffusion model with LoRA adapters and a diffusion self-consistency loss to generate high-fidelity faces from the predicted identity vectors.

In this framework, $\hat{z}_{arc} \in \mathrm{R}^{512}$ is the only explicit identity vector that flows across modalities, and it is always normalized on the ArcFace unit hypersphere, i.e., $\| \hat{z}_{arc} \|_2 = 1$. This concentrates cross-modal instability into a measurable, testable geometric space (e.g., cosine or angular distance), while the generative backend reuses the Stable Diffusion latent denoising backbone already pretrained in Arc2Face. The backbone architecture and weights remain frozen; we only insert lightweight LoRA adapters along the conditioning pathways, which provides a natural interface for the diffusion self-consistency alignment mechanism detailed in Section 4.2.

4.2. Training Objectives and Optimization

During training, we jointly optimize the speech-to-identity mapping network g_ϕ and the LoRA parameters ψ_{txt}, ψ_{unet} injected into the Arc2Face text encoder and UNet, while keeping both the Arc2Face backbone and the ArcFace encoder frozen. The overall objective consists of two components: a cross-modal identity alignment loss and an Arc2Face-based SDS-style diffusion self-consistency loss.

Given a paired sample (a, x, y), where a is the speech segment, x is the face image, and y is the identity label, we first extract the image-side identity embedding with ArcFace

$$z_{arc} = f_{arc}(x) \in \mathrm{R}^{512}, \| z_{arc} \|_2 = 1 \tag{6}$$

then obtain the speech-side predicted identity embedding

$$\hat{z}_{arc} = g_\phi(f_{spk}(a)) \tag{7}$$

which is ℓ_2-normalized onto the unit hypersphere.

Cross-modal identity alignment loss. To ensure that $\hat{z}_{arc}$ has clear identity separation and a well-structured cross-modal pairing under the ArcFace metric, we combine an Additive Margin Softmax (AM-Softmax) loss [22] with an InfoNCE contrastive loss [30].

On the one hand, we attach an AM-Softmax classification head to $\hat{z}_{arc}$. With angular margin m and scale s, the target-class logit $\cos\theta_y$ is replaced by $s(\cos\theta_y - m)$, while the logits for non-target classes remain $s\cos\theta_j$. This yields the identity classification loss

$$L_{AM} = -log\frac{\exp(s(\cos\theta_y - m))}{\exp(s(\cos\theta_y - m)) + \sum_{j\neq y}\exp(s(\cos\theta_j))} \tag{8}$$

Large-margin softmax variants of this form have been widely shown to improve the discriminativeness of embeddings in face and speaker recognition.

On the other hand, we apply an InfoNCE contrastive loss over a mini-batch of paired speech–face embeddings $\left(\hat{z}_{arc}^{(i)}, z_{arc}^{(i)}\right)$. Treating each pair as a positive pair and all other combinations within the batch as negatives, with batch size B and temperature τ, the InfoNCE loss is

$$L_{InfoNCE} = -\frac{1}{B}log\frac{\exp\left(\cos\left(\hat{z}_{arc}^{(i)}, z_{arc}^{(i)}\right)/\tau\right)}{\sum_{j=1}^{B}\exp\left(\cos\left(\hat{z}_{arc}^{(i)}, z_{arc}^{(j)}\right)/\tau\right)} \tag{9}$$

In Equation (9), we implement InfoNCE with in-batch negatives for computational efficiency and stable optimization. While hard-negative mining is a potential enhancement for contrastive objectives, in identity alignment it may introduce false negatives: acoustically similar speakers or embedding noise can produce near-positive pairs that are mistakenly treated as negatives, which can harm intra-class compactness and destabilize training. In our framework, strong inter-class separability is primarily enforced by the margin-based AM-Softmax loss, whereas InfoNCE acts as an auxiliary instance-level constraint to encourage correct cross-modal pairing within a batch. Therefore, we adopt in-batch negatives as a robust baseline in this work, and leave more sophisticated negative mining strategies (e.g., queue-based negatives or hard-negative selection with false-negative mitigation) for future investigation.

The identity alignment objective is then given by the weighted sum

$$L_{id} = \lambda_{AM}L_{AM} + \lambda_{Info}L_{InfoNCE} \tag{10}$$

where λ_{AM} and λ_{Info} control the relative contributions of the two terms. Although ArcFace defines the target identity hypersphere used for supervision, AM-Softmax is a margin-based objective applied to the speech-induced identity embeddings to explicitly enforce inter-class angular margins on that hypersphere. The InfoNCE term, in contrast, provides an instance-level cross-modal matching constraint by aligning each speech embedding with its paired face embedding relative to other samples in the batch. We therefore combine AM-Softmax and InfoNCE to jointly encourage (i) global class separability and (ii) correct pairwise alignment structure, rather than using either objective alone.

Diffusion self-consistency loss. Although $\hat{z}_{arc}$ learned under L_{id} is aligned with the target identity in the metric space, it may still fall outside the training distribution on which Arc2Face was pretrained, making it difficult to decode into high-quality, natural face images. To further enforce that $\hat{z}_{arc}$ is both identity-consistent and generatable, we construct an SDS-style diffusion self-consistency loss based on the denoising process of Arc2Face, effectively using the model as its own quality critic and forming a "generate–analyze–regenerate" loop [28,29].

Specifically, for a given predicted identity embedding $\hat{z}_{arc}$, we first obtain the identity condition via the Arc2Face projection head

$$c_{gen} = P(\hat{z}_{arc}) \tag{11}$$

In the latent space, we sample an initial noise $z_T \sim N(0, I)$ and follow the forward diffusion process to obtain an intermediate latent state z_t. Conditioned on c_{gen} and timestep t, the LoRA-augmented Arc2Face UNet $\epsilon_\theta(\cdot, c, t)$ predicts the noise and performs reverse denoising to recover z_0 and its corresponding generated image I_{gen}.

The key "self-consistency" step is to extract the identity implied by the generated image. We feed I_{gen} into the frozen ArcFace encoder to obtain

$$z_{self} = f_{arc}(I_{gen}) \tag{12}$$

and pass it through the same projection head to obtain a self-induced condition

$$c_{self} = P\left(z_{self}\right) \tag{13}$$

Intuitively, if $\hat{z}_{arc}$ is an ideal identity condition, then the image generated under c_{gen} should imply (via ArcFace and P(·)) a condition c_{self} that leads to nearly identical denoising behavior on the same noisy latent z_t. Based on this intuition, we define the diffusion self-consistency loss as the difference between noise predictions under the two conditions:

$$L_{self} = E_{t,\epsilon}\left[w(t)\left\|\epsilon_\theta(z_t, c_{gen}, t) - \epsilon_\theta\left(z_t, c_{self}, t\right)\right\|_2^2\right] \tag{14}$$

where $w(t)$ is a timestep-dependent weighting function, z_t is the latent at timestep t in the forward process, and $\epsilon_\theta(\cdot, c, t)$ denotes the UNet's noise prediction given condition c and timestep t. Note that the proposed diffusion self-consistency regularizes denoising residual consistency under matched stochastic conditions rather than minimizing pixel-space reconstruction errors. Therefore, it does not explicitly drive images toward averaged appearances. Instead, it encourages the predicted identity condition to remain within the high-probability conditioning region of the pretrained generator while keeping the backbone frozen. Since the final images are still produced via stochastic diffusion sampling, visual variability is preserved.

In implementation, we apply a stop-gradient operation to c_{self} and freeze the entire Arc2Face backbone, the ArcFace encoder, and the path used to compute c_{self}. Consequently, gradients from L_{self} flow only through c_{gen} back to its sources, i.e., the speech mapping network g_ϕ and the LoRA parameters ψ_{txt}, ψ_{unet}. This encourages $\hat{z}_{arc}$ to contract towards regions of the identity space where Arc2Face has high generative probability, while preserving discriminative identity structure.

The final training objective of Vox2Face is

$$L_{total} = \lambda_{id} L_{id} + \lambda_{self} L_{self} \tag{15}$$

where λ_{id} controls the strength of identity alignment and λ_{self} controls the weight of the diffusion prior constraint.

We adopt a simple two-stage training scheme. In Stage I (identity pre-alignment), we set $\lambda_{self} = 0$ and train only g_ϕ (and optionally the top layers of the speech encoder) with L_{id}, so that speech-predicted identity vectors already exhibit clear inter-class margins and cross-modal pairing structure in the ArcFace space. In Stage II (joint optimization with diffusion self-consistency), we load the pretrained Arc2Face weights, inject LoRA modules into its text encoder and UNet, and jointly optimize g_ϕ, ψ_{txt}, and ψ_{unet} under both L_{id} and L_{self}. In this stage, AM-Softmax and InfoNCE continue to enforce identity consistency between speech and faces, while the SDS-style self-consistency loss, together with lightweight LoRA adaptation, further aligns the speech-driven identity distribution

with the Arc2Face generative prior—achieving improvements in both identity fidelity and image quality without significantly altering the diffusion backbone.

5. Experiments

5.1. Dataset and Evaluation Protocol

Speech Fusion to Face (SF2F) [31]. HQ-VoxCeleb is constructed by re-collecting and refining high-quality face images for the identity intersection of the VoxCeleb audio–visual corpus, providing paired speech–face data with reliable identity labels under a controlled, high-resolution evaluation protocol. Starting from VoxCeleb identities, the SF2F authors apply face detection, alignment, segmentation, and reconstruction to better control pose, illumination, expression, and background, so that speech-to-face models can focus on speech–identity association rather than low-level denoising or image normalization. We note that, to the best of our knowledge, HQ-VoxCeleb is currently the only publicly available benchmark that explicitly supports high-resolution, identity-preserving speech-to-face generation with curated face images and identity-level splits; therefore, our empirical evaluation is scoped to the HQ-VoxCeleb/SF2F setting. According to the public statistics in SF2F, HQ-VoxCeleb contains 3638 identities, 8028 high-quality face images, and 609,700 speech segments, split by identity into 2890/370/378 identities for training/validation/test, with an average image resolution of about 505 × 505 pixels. The data split is summarized in Table 1.

Table 1. HQ-VoxCeleb data setting.

	Face Images	Voice Segments	Identities
Train	6375	478,009	2890
Validation	859	63,337	370
Test	794	68,354	378
Total	8028	609,700	3638

During both training and evaluation, we use HQ-VoxCeleb only as a source of paired speech–face supervision: each sample consists of a speech segment and a frontal, neutral-expression face image of the same speaker. No textual labels, attribute annotations, or external reference faces are used, so evaluation focuses on identity consistency and visual quality rather than auxiliary multimodal cues. For audio preprocessing, we follow the SF2F protocol to perform clip-level segmentation: raw 16 kHz waveforms are segmented into 1.25 s audio clips with 50% overlap. For each clip, we compute log-Mel spectrograms using conventional short-time analysis with a 25 ms frame length (as in standard speaker/audio feature extraction), and then apply the standard AST preprocessing steps before feeding the resulting features to the speaker encoder. On the image side, we use the corresponding HQ face image for each identity; if multiple images exist for one identity, we randomly sample one during training.

Evaluation protocol and metrics. To enable a fair comparison with SF2F, we evaluate Vox2Face on the HQ-VoxCeleb test split using three complementary groups of metrics: identity similarity, cross-modal retrieval, and image quality. Since speech-driven face generation is inherently one-to-many, our evaluation prioritizes identity faithfulness and overall face realism/recognizability under the identity-preserving setting.

For identity similarity, we extract features from both real faces $\hat{x}$ and generated faces x using a pretrained ArcFace network, and compute cosine similarity (Cosine), which measures how close the generated identity is to the ground truth in a shared embedding space, and l_2 distance (L2), which provides a complementary view of absolute embedding

deviation. For speech-to-face retrieval, we follow the SF2F protocol: ArcFace features of generated faces are used as queries over all real faces in the test set, and we report Recall@K (R@5 and R@10), i.e., the fraction of test identities whose ground-truth face appears among the top-5/top-10 nearest neighbors. For image quality, we adopt the VGGFace Score (VFS) from SF2F, which feeds generated images into a pretrained VGGFace classifier to obtain identity posteriors and then computes an exponentiated mutual-information-style score over all samples. Intuitively, VFS plays a role analogous to Inception Score but is tailored to face generation, jointly reflecting recognizability and overall perceptual quality. Together, ArcFace-based Cosine/L2 and R@5/R@10, as well as VGGFace-based VFS, provide a comprehensive assessment of Vox2Face in terms of identity fidelity, retrieval performance, and visual realism.

5.2. Training Setup

Speaker encoder and audio preprocessing. On the speech side, we closely follow the configuration used in Face from Voice [32] and adopt the Audio Spectrogram Transformer (AST) as our speaker encoder. This choice is motivated by our objective to obtain a robust, transferable speech representation for downstream speech-to-face identity mapping, rather than to maximize standalone speaker-verification accuracy. All audio clips are converted to mono and resampled to 16 kHz. We then apply pre-emphasis, frame the signal with a sliding window, and compute 128-bin log-Mel spectrograms, using the same time–frequency cropping and normalization settings as in the original AST pipeline. AST is initialized from the AudioSet-pretrained weights released by the authors, and we do not modify its architecture or hyperparameters. The speaker embedding $v_{\text{spk}} \in \mathrm{R}^{d_{\text{spk}}}$ is obtained from the [CLS] token of AST.

The embedding v_{spk} is then fed into a three-layer MLP g_ϕ (hidden width 1024, SiLU activation, LayerNorm, dropout 0.1). The final linear layer maps it to a 512 dimensions followed by ℓ_2 normalization, yielding a unit-norm identity vector $\hat{z}_{arc} \in \mathrm{R}^{512}$. On top of $\hat{z}_{arc}$, we apply the AM-Softmax identity classification loss together with the batch-wise InfoNCE contrastive loss described in Section 4.2, so that speech-induced identity embeddings are aligned with real face embeddings under the spherical geometry of ArcFace while preserving clear inter-class angular margins and cross-modal matching structure.

Conditioning interface (speech vs. text). Vox2Face does not condition the diffusion model on speech tokens or text-like prompts. Instead, the speech encoder and mapping network predict an ArcFace identity embedding $\hat{f}_{arc}$, and the generator is conditioned on this identity vector (Equation (5)). This preserves the Arc2Face-style identity-conditioning interface, where generation is controlled by identity embeddings rather than linguistic content.

Generator and conditional injection (Arc2Face@SD-1.5). The generative backbone builds on the identity-conditioned interface provided by Arc2Face, whose base model is a latent diffusion model derived from Stable Diffusion v1.5. Arc2Face takes an ArcFace identity vector as its only conditioning signal: it uses a fixed pseudo-text prompt (e.g., "a photo of <id> person"), represents <id> by a dedicated placeholder token, replaces that token's embedding with the ArcFace identity vector, and feeds the resulting sequence into a finetuned CLIP ViT-L/14 text encoder to obtain a 77-token CLIP sequence, which is injected into the UNet denoiser via cross-attention. We keep this ID-conditioned architecture intact and introduce LoRA adapters only for parameter-efficient adaptation, without changing the conditioning interface.

Stable Diffusion v1.5 uses a variational autoencoder (VAE) to bridge pixel space and latent space: the encoder applies an $8\times$ downsampling factor, mapping an $H \times W \times 3$ RGB image to an $(H/8) \times (W/8) \times 4$ latent tensor. Thus, a 512×512 image corresponds to a $64 \times 64 \times 4$ latent, which substantially reduces per-step computational cost. During

training, the RGB inputs to the VAE encoder are the ground-truth face images from HQ-VoxCeleb, which we encode into latents for latent-space diffusion training; during inference, and generation starts from a Gaussian latent that is denoised under ID conditioning and finally decoded to RGB by the VAE decoder. In our implementation, we directly reuse the official Arc2Face repository and the corresponding Hugging Face weights, keeping the ID-conditioned architecture intact and only adding LoRA adaptation modules together with our speech-side mapping network.

Training and inference settings. We adopt a two-stage training strategy. In the identity pre-alignment stage, ArcFace and Arc2Face (including the CLIP text encoder, UNet, and VAE) are fully frozen, and we train only g_ϕ and the top layers of the speaker encoder. We use AdamW as the optimizer with an initial learning rate of 1×10^{-4}, weight decay of 0.01, batch size of 128, and roughly 100k training steps. The loss function is the identity alignment loss L_{id} defined in Section 4.2 with cosine learning-rate decay.

In the diffusion self-consistency stage, we load the pretrained Arc2Face weights and insert LoRA modules into the attention projections of both the text encoder and the UNet (i.e., low-rank adapters on the projection matrices in attention layers, such as the $Q/K/V$ and output projections). The default LoRA rank is $r = 16$ with scaling factor 16, and we optimize only the LoRA parameters ψ_{txt}, ψ_{unet} together with g_ϕ, keeping the Arc2Face backbone frozen. We again use AdamW, with a learning rate about 1×10^{-6} and a batch size of 16. In addition to L_{id}, we include the diffusion self-consistency loss L_{self} (Section 4.2). To improve numerical stability and convergence, we apply timestep truncation, gradient clipping, and weight warm-up.

Rank ablation. To justify the LoRA capacity choice, we additionally evaluate four rank settings $r \in \{4, 8, 16, 32\}$ with all other configurations fixed, and report the results in Section 5.3.

Pretrained models and weight sources. We use publicly released pretrained checkpoints for all backbone components. Specifically, the Stable Diffusion v1.5 backbone and the ArcFace-related identity components used by Arc2Face are obtained from the Hugging Face model release FoivosPar/Arc2Face. The audio encoder is initialized from the Hugging Face Transformers release of the Audio Spectrogram Transformer (AST). For VGGFace-based evaluation, we use the VGGFace weights provided by the public release of BAI-Yeqi/SF2F_PyTorch.

At inference, we follow the sampling configuration recommended by Arc2Face and used in its official implementation: we adopt DPM-Solver++ (DPMSolverMultistepScheduler) as the diffusion sampler on top of Stable Diffusion v1.5. Starting from a Gaussian latent, DPM-Solver++ iteratively updates the latent using the UNet's noise predictions for 25 steps in latent space, after which the VAE decoder produces 512 × 512 face images. The inference cost scales approximately linearly with the number of sampling steps S(i.e., $O(S \cdot C_{UNet(latent)})$). All experiments are conducted on an NVIDIA A800 80GB GPU (NVIDIA, Santa Clara, CA, USA). Our implementation is based on PyTorch 3.10 and the Stable Diffusion pipeline from the Hugging Face diffusers library [5,33]; we fix random seeds and enable mixed-precision training and inference. Audio segmentation and image loading for HQ-VoxCeleb strictly follow Section 5.1, ensuring reproducible evaluation and fair comparison. On this setup, Stage-I (step 0–10,000) takes 8.4 h, Stage-II (step 10,000–30,000) takes 34.2 h, and inference with 25-step DPM-Solver++ sampling takes 17–18 s per batch of 4 generated images.

5.3. Quantitative Analysis

We first evaluate Vox2Face on the HQ-VoxCeleb test set against several representative baselines, including [9] instantiated via the reproducible PyTorch adaptation by Pietkun

et al. [13], the autoencoder-based Speech2Face (PyTorch reimplementation) and its AST-based variant AST_165 [13], Voice2Face [11], and the SF2F model [31]. All methods share the same train/test split and preprocessing, and we follow the evaluation protocol and metric definitions in Section 5.1 to ensure a consistent and fair comparison.

Table 2 summarizes the cosine similarity and L2 distance computed in the ArcFace embedding space, together with retrieval metrics (R@5/R@10) based on ArcFace features, and VFS computed using VGGFace. In terms of identity similarity, Vox2Face achieves the best cosine similarity of 0.322, corresponding to ≈9% relative improvement over the strongest baseline SF2F (0.295). At the same time, it attains the lowest l2 distance of 1.09, outperforming SF2F (1.14) and other baselines (1.09–1.26). These results indicate that, in the ArcFace identity space, Vox2Face's generated faces lie closer to the ground-truth identity embeddings, reflecting more accurate cross-modal alignment from speech to identity vectors.

Table 2. Quantitative comparison on the HQ-VoxCeleb test set.

Methods	Similarity		Retrieval		Quality
	Cosine	L2	R@5	R@10	VFS
Speech2face [13]	0.213	1.20	4.6	8.7	13.43
AST_165 [13]	0.216	1.17	5.7	10.5	13.78
Voice2Face [11]	0.259	1.26	8.7	12.3	15.03
SF2F [31]	0.295	1.14	14.6	29.8	18.82
Vox2Face (ours)	**0.322**	**1.09**	**17.3**	**32.1**	**23.21**

In retrieval performance, Vox2Face again delivers the best results: R@5 and R@10 reach 17.3% and 32.1%, improving upon SF2F (14.6%/29.8%) by 2.7 and 2.3 absolute points, with even larger gains over earlier Speech2Face and Voice2Face variants. Thus, when generated faces are used to retrieve the corresponding speaker from a gallery, Vox2Face more reliably ranks the true identity among the top-K results, providing further evidence that the identity-centric design strengthens cross-modal identity consistency.

For image quality, Vox2Face attains the highest VFS of 23.21, significantly higher than SF2F (18.82) and the remaining baselines (13.43–15.03). Since VFS jointly reflects image sharpness, facial structural plausibility, and identity discriminability, these results suggest that combining Arc2Face as a high-quality face prior with diffusion self-consistency and LoRA adaptation allows Vox2Face to produce faces that are both closer to the true identity and visually more stable and natural. Overall, across identity similarity, retrieval accuracy, and perceptual quality, we observe a clear trend: as models evolve from Speech2Face and Voice2Face to SF2F, all three dimensions steadily improve; on top of this progression, Vox2Face further advances all metrics by unifying the ArcFace identity space, reusing an identity-conditioned diffusion prior, and introducing diffusion self-consistency optimization, quantitatively validating our "align identity space first, then generate via a strong identity prior" design.

Overall, compared with SF2F, Vox2Face improves Cosine by +0.027, R@5 by +2.7 points, R@10 by +2.3 points, and VFS by +4.39, demonstrating consistent gains in identity fidelity, retrieval accuracy, and perceptual quality under the same HQ-VoxCeleb protocol.

To quantitatively justify each component of Vox2Face, we first study LoRA adaptation capacity via the rank r (Table 3). We then report a minimal ablation matrix to isolate the identity-space alignment design and diffusion self-consistency (DSC) as an additional regularizer beyond L_{id} (Table 4). All ablations are evaluated under the same HQ-VoxCeleb/SF2F protocol and metrics.

Table 3. LoRA capacity. LoRA injects low-rank trainable updates while freezing the backbone; therefore, the rank r directly controls adaptation capacity with a small number of trainable parameters, motivating the sensitivity study.

LoRA Rank	Similarity		Retrieval		Quality
r	Cosine	L2	R@5	R@10	VFS
4	0.314	1.14	15.5	27.3	22.92
8	0.317	1.12	16.6	30.2	23.09
16	0.322	**1.09**	**17.3**	**32.1**	23.21
32	**0.323**	1.09	17.0	31.7	**23.23**

Table 4. Component attribution. Table 4 isolates the effect of DSC by comparing the L_{id}-only variant (w/o DSC) with the full model, showing consistent gains across similarity, retrieval, and VFS, which supports DSC as an independent contributor beyond identity-space alignment.

Settings	Similarity		Retrieval		Quality
	Cosine	L2	R@5	R@10	VFS
w/o ArcFace	0.242	1.17	13.4	25.6	19.65
w/CosReg	0.317	1.10	12.9	22.5	21.64
w/o DSC	0.296	1.12	16.2	30.4	21.87
full Vox2Face	**0.322**	**1.09**	**17.3**	**32.1**	**23.21**

We evaluate the impact of LoRA capacity by varying the rank $r \in \{4, 8, 16, 32\}$ while keeping all other configurations fixed. LoRA adapters are inserted into the attention layers of both the text encoder and the UNet, and only the LoRA parameters together with the speech-side mapping network are optimized in Stage II. We report identity fidelity (Cosine, L2, and retrieval R@K) and image quality (VFS) under the same evaluation protocol as the main comparison.

As shown in Table 3, increasing the LoRA rank generally improves performance up to a certain point, reflecting the larger adaptation capacity available for aligning speech-derived identity conditions with the pretrained diffusion prior. However, the gains saturate when r becomes large, suggesting that overly high rank yields diminishing returns under the current training budget and data scale. Overall, $r = 16$ offers a favorable balance between identity fidelity and visual quality, and we adopt it as the default setting in all other experiments.

To assess the contribution of each key component, we conduct ablation studies on the HQ-VoxCeleb test set. Unless otherwise specified, all variants share the same train/validation split and preprocessing as in the main experiments, and all models use the same AST configuration as *Face from Voice* on the speech side, so performance differences can be attributed to identity-space modeling and the generative backend rather than the speech encoder.

We first examine whether explicitly aligning speech embeddings to the ArcFace identity space is necessary. The w/o ArcFace variant keeps the AST encoder and Arc2Face generator but removes g_ϕ and ArcFace distillation; instead, a shallow MLP maps speech features directly into the Arc2Face conditioning space, trained with a simple l_2 regression and reconstruction loss. Compared with full Vox2Face, this variant suffers a clear drop in cosine similarity (0.242 vs. 0.322), and R@5/R@10 fall to 13.4%/25.6%. This indicates that "directly driving the diffusion model from speech" struggles to learn stable cross-modal identity relationships from limited paired data, whereas ArcFace alignment constrains

the identity geometry on the unit hypersphere and aligns it with the class-wise angular structure of a face recognition model, leading to much tighter identity-related metrics.

We then analyze the impact of the cross-modal alignment loss design. The w/CosReg variant uses only a simple cosine regression loss between $\hat{z}_{arc}$ and z_{arc}. It achieves cosine and ℓ_2 scores close to the full model (0.317/1.10 vs. 0.322/1.09), but its retrieval performance is clearly worse (R@5/R@10 = 12.9%/22.5%). This suggests that pointwise regression can approximate the target embedding on average but fails to enforce clear class boundaries and pairing structure at the batch level, which are essential for robust cross-modal retrieval.

Finally, we isolate the effect of Diffusion Self-Consistency (DSC). The w/o DSC variant, after aligning to the ArcFace space, trains g_ϕ and the LoRA modules using only L_{id}; the generator thus acts as a passive decoder without additional constraints on the speech-side identity representation. As shown in Table 4, removing DSC yields consistent degradations across metrics: Cosine drops from 0.322 to 0.296, L2 increases from 1.09 to 1.12, R@5 decreases from 17.3 to 16.2, R@10 decreases from 32.1 to 30.4, and VFS decreases from 23.21 to 21.87. These results indicate that DSC provides a useful corrective signal from the diffusion prior, improving both identity fidelity and image quality beyond ArcFace-space alignment.

Overall, the ablations show that (i) aligning to the ArcFace identity space is the key prerequisite for performance gains, (ii) combining AM-Softmax with InfoNCE is clearly superior to simple cosine regression for identity metric learning and cross-modal matching, and (iii) adding an SDS-style diffusion self-consistency constraint further pulls speech-induced identity vectors toward the high-probability generative manifold of Arc2Face, yielding consistent improvements in identity similarity, retrieval metrics, and VFS, while LoRA-based adaptation avoids the cost and instability of fully finetuning the diffusion backbone.

To make the optimization behavior more transparent, we provide learning-curve visualizations for both training stages, as shown in Figure 2. We plot the training losses (L_{id} in Stage I; L_{id} and L_{self} in Stage II) over training steps, illustrating convergence and stability of the proposed approach.

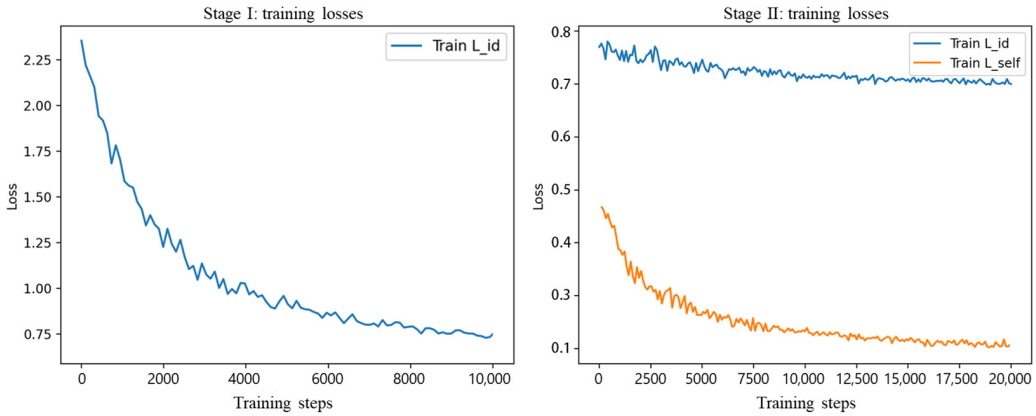

Figure 2. Training curves of Vox2Face. Left: Stage I identity pre-alignment (training L_{id}). Right: Stage II diffusion self-consistency with LoRA (training L_{id} and L_{self}). Curves are reported over training steps with the same train split as in the main experiments.

5.4. Qualitative Analysis

For visual consistency, we apply a simple post-processing step to the generated images for visualization only. Raw Vox2Face outputs are face images rendered with a simple background. Following SF2F, we run the same face detection and foreground segmentation pipeline to extract and re-align the facial region onto a uniform white background. This operation is used solely for presentation; all identity feature extraction and quantitative evaluations are performed on the original high-resolution outputs without any

post-processing. Unless otherwise stated, all methods shown in Figure 3 are rendered using the same face-cropping and white-background pipeline.

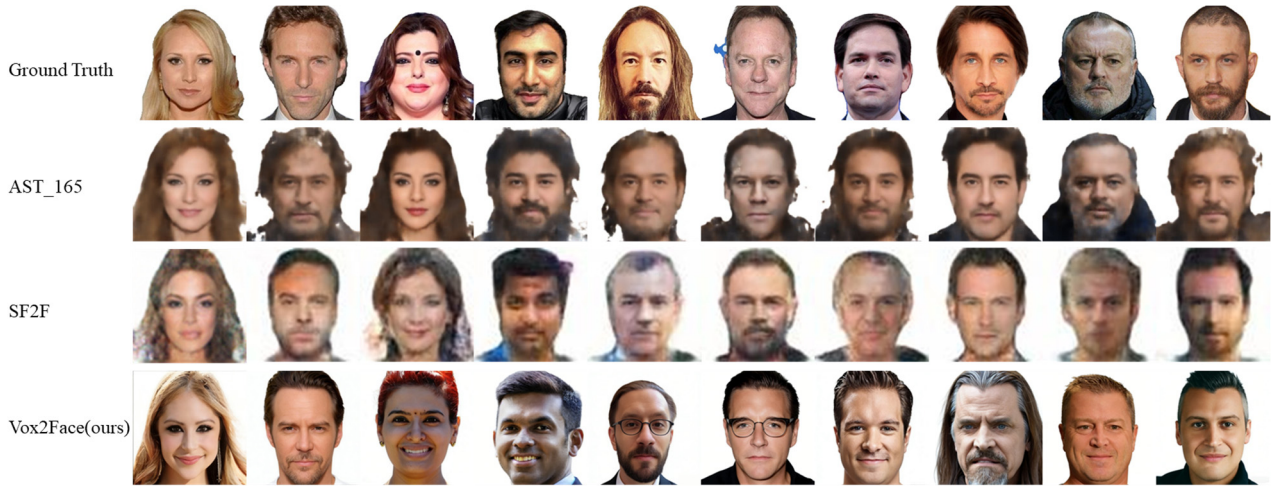

Figure 3. Example results on HQ-VoxCeleb. For each test utterance, we show the ground-truth HQ face and faces generated by Face from Voice, SF2F, and the proposed Vox2Face. Overall, Vox2Face tends to produce faces that are visually sharper and more identity-consistent under the same speech input, while baseline methods more often exhibit reduced inter-speaker distinctiveness. All images in the figure are rendered with the same visualization pipeline described above; quantitative results are computed on the original outputs.

As shown in Figure 3, earlier approaches based on autoencoder/GAN-style generation can capture coarse attributes such as gender and approximate age that are broadly consistent with the input speech, but their outputs often suffer from blur and artifacts: facial contours are underdeveloped, fine details are missing, and some samples exhibit unnatural textures or structural distortions. This behavior is consistent with the limited strength of the identity prior in these paradigms.

SF2F further improves perceptual quality: most of its outputs have more complete facial structure with more natural skin tone and illumination. However, SF2F still exhibits a non-negligible "average face" effect: faces of different speakers often share similar global shape and facial style, and fine-grained identity cues such as hairstyle, beard, or eyeglasses are only weakly expressed. As a result, SF2F tends to adopt a "safe but less personalized" generation strategy, trading some individuality to avoid artifacts and maintain stable visual quality.

By contrast, Vox2Face exhibits stronger identity specificity along several dimensions. In terms of identity-related attributes, Vox2Face produces faces whose gender, age group, skin tone, and coarse ethnicity are more consistent with the true speakers. This is especially evident for very young and elderly speakers, where changes in facial texture, wrinkle patterns, and overall face shape are more pronounced, avoiding the tendency to "flatten" different age groups into a neutral adult appearance. Because speech embeddings are explicitly aligned to the ArcFace identity space and we reuse the Arc2Face identity-conditioned diffusion prior, Vox2Face can retain richer individual differences while preserving identity consistency; hairstyle silhouettes, forehead and jawline shape, facial hair, and eyeglasses are often closer to the reference faces rather than being over-smoothed away.

In terms of visual quality, Vox2Face produces images that are overall sharper and more stable, with natural facial structure and almost no obvious texture tearing or local distortions. Even when the background is complex or illumination varies, the generated faces remain structurally coherent, which is consistent with the improvements observed in VFS. At the same time, we also observe limitations under extreme conditions: when

input speech segments are very short or heavily corrupted by noise and reverberation, the model sometimes falls back to a more conservative neutral-looking face, with hairstyle and local details regressing toward high-frequency patterns in the training distribution. This suggests that, when the speech signal carries insufficient identity information, outputs are still influenced by the prior distribution encoded in the pretrained identity model.

Overall, the qualitative findings align with the quantitative results: under the same speech input, Vox2Face consistently improves over existing methods in both identity consistency and visual quality. This supports our central design choice—unified ArcFace identity space + identity-conditioned diffusion prior + diffusion self-consistency optimization—as an effective way to enhance identity fidelity and visual realism in speech-driven face generation without retraining a large diffusion backbone.

Although Vox2Face improves identity fidelity and visual quality on average, we observe a subset of test cases where the generated faces deviate from the ground truth in systematic ways. Figure 4 summarizes representative failure patterns on HQ-VoxCeleb. These failure cases reflect the inherent ambiguity of speech-to-face generation: speech does not uniquely specify all facial attributes, and some non-identity factors (e.g., accessories and illumination) are partially determined by the generative prior and training distribution.

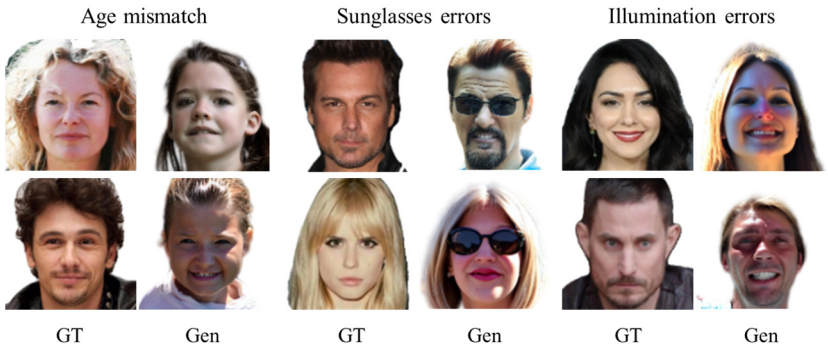

Figure 4. Failure cases and error taxonomy on HQ-VoxCeleb. For each example, the ground-truth face (GT) is shown on the left and the generated face (Gen) on the right. From left to right, we illustrate three representative failure types observed on the test set: (i) age mismatch, (ii) sunglasses-related errors (hallucinated or inconsistent sunglasses rendering), and (iii) illumination/color-cast errors (unnatural lighting or color shifts). All images are rendered using the same visualization pipeline as Figure 3.

6. Conclusions

This paper presented Vox2Face, a speech-driven face generation framework that shifts the focus from direct speech-to-image mapping to identity-centric modeling. By distilling AST-based speaker embeddings into the ArcFace hyperspherical space and reusing Arc2Face as a frozen identity-conditioned diffusion backbone with lightweight LoRA adapters, Vox2Face reformulates speech-to-face generation as identity similarity modeling in a unified metric space. On HQ-VoxCeleb, this design yields consistently better identity preservation, speech-to-face retrieval performance, and perceptual quality than representative baselines, while retaining inference efficiency comparable to the original Arc2Face model.

We further introduce a diffusion self-consistency objective that acts as a regularizer on the conditional denoising dynamics, using denoising residuals to encourage speech-derived identity vectors to remain consistent with both the ground-truth ArcFace embeddings and identity regions that are likely under the pretrained generator's prior. Looking ahead, we will strengthen contrastive alignment under potential identity imbalance by adopting more robust negative-sampling strategies, including queue-based dictionaries and hard-

negative mining with explicit false-negative mitigation, while maintaining training stability. We also plan to extend Vox2Face from identity-preserving face synthesis toward more controllable talking-head generation by incorporating explicit controls for expression, pose, and linguistic content, and to improve robustness under cross-lingual and noisy conditions. To better characterize diversity and rule out potential over-smoothing in this inherently one-to-many setting, we will add LPIPS-based diversity analyses across multiple stochastic generations per utterance. Finally, we will integrate interpretability, access control, and de-identification mechanisms to better address the privacy and ethical risks associated with reconstructing faces from speech.

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Data Availability Statement: The data presented in this study are openly available in the VoxCeleb repository and the HQ-VoxCeleb dataset, with documentation and access information provided at https://www.robots.ox.ac.uk/~vgg/data/voxceleb/ (accessed on 10 August 2025) and https://github.com/BAI-Yeqi/SF2F_PyTorch (accessed on 10 August 2025).

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